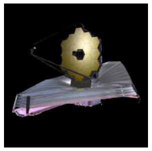

11164 - Characterizing the Accreting Protoplanet Embedded in the Protoplantary Disk

Cycle: 5, Proposal Category: GO

INVESTIGATORS

Name	*Institution*
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Dr. Michihiro Takami (CoI)	Academia Sinica, Institute of Astronomy and Astrophysics

OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
Observation Folder				
	1		NIRSpec IFU Spectroscopy	(1) IRAS-19205-0746

ABSTRACT

The planetary growth rate, specifically the mass accretion rate, is a key parameter in planet formation, as it ultimately determines a planet's final mass. Traditionally, accretion rates have been estimated using hydrogen emission lines based on the accretion flow model, originally developed for stars. However, Aoyama et al. recently proposed the accretion shock model as an alternative mechanism for hydrogen line emission. For planetary-mass objects, the shock model predicts accretion rates up to an order of magnitude higher than the flow model, making it crucial to determine which mechanism dominates.

We propose a joint JWST-HST observing program of the protoplanet WISPIT 2 b to distinguish between the two models using hydrogen line ratios. For the first time, this study will enable more accurate estimates of accretion rates in accreting planets embedded within protoplanetary disks. Our pilot observations pave the way for applying these models to characterize future discoveries with JWST and the ELTs.

OBSERVING DESCRIPTION

This is a joint JWST-HST observing program. The primary observatory is JWST. We request contemporaneous JWST and HST observations. According to the visit planner of JWST and HST, our target can be simultaneously observed with JWST and HST from April to May in 2027.

We request to use NIRSpec IFU with the G140H/F100LP setting. The parameters in detector setup is as follows: Subarray of full, Readout pattern of NRSIRS2RAPID, Groups per integration of 6, Integrations per exposure of 18, Total Dithers of 4. Total exposure time in ETC is Total exposure time is 02:02:33 (7352.80 s). We also request 4-point-nod to mitigate cosmic rays and to subtract backgrounds.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	IRAS-19205-0746	RA: 19 23 17.0332 (290.8209717d) Dec: -07 40 55.08 (-7.68197d) Equinox: J2000	Proper Motion RA: 6.308 mas/yr Proper Motion Dec: -27.137999904880417 mas/yr Parallax: 0.0074649" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[Exoplanet Systems, Exoplanets]
Extended=NO

Observation

Proposal 11164, Observation 1

Wed Jun 03 18:00:11 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
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Category=Star
Description=[Exoplanet Systems, Exoplanets]
Extended=NO

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	CYCLING	MEDIUM	1	12	

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G140H/F100LP	NRSIRS2RAPID	6	6	false	true	NONE	12	72	7352.801	273320.1